## a Data Acquisition & Programmatic Verification Workflow

**1. Ingestion Layer**

- Official FIFA Match Center (104 Matches, TSG Reports)
- Sofascore API Endpoints (601 Events, Team Telemetry)
- Transfermarkt Squad Roster (1,248 Player Valuations)
- Geographic Elevation Surveys (16 Host Venues)

**2. Resolution & Anomaly Layer**

- Canonical entity disambiguation & diacritic stripping
- 27 Raw ingestion discrepancies logged & resolved (1.00% rate)
- Deterministic integer PK assignment & UTC timestamps

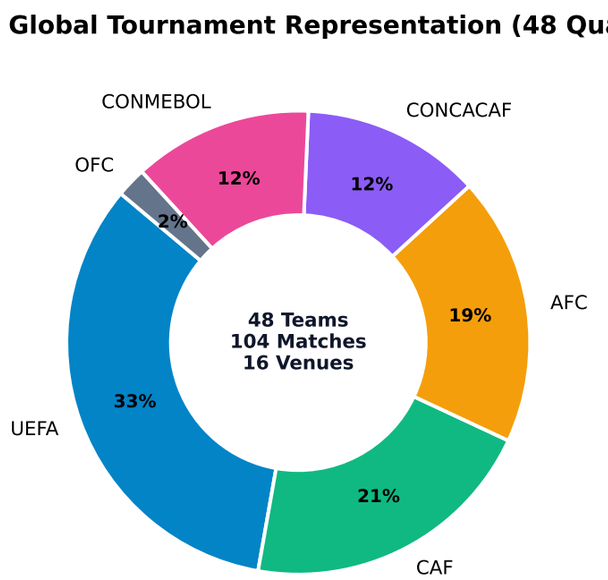
**3. 3NF Database Compilation**

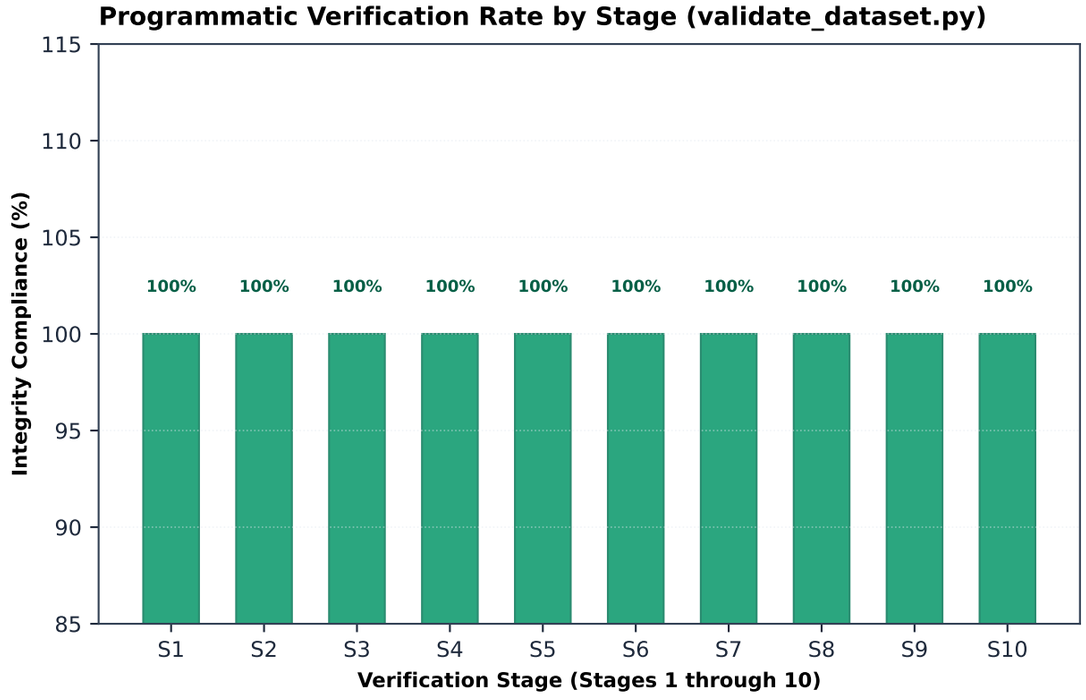
- 12 Normalized Relational Entities (teams, matches, lineups, ...)
- Multi-Format Parity: RFC 4180 CSV, Snappy Parquet, SQLite DB
- Enforced Constraints: PRAGMA foreign_keys = ON (0 Violations)

**4. Scientific Verification Suite**

- Stages 1–2: Entity Counts & Primary Key Uniqueness
- Stages 3–5: Referential Integrity & Match Status Logic
- Stages 6–8: 26-Squad Lineups & 308 Goals Reconciled
- Stages 9–10: Neutral Officials & Multi-Engine Parity

## b

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